BARI 공동연구

물리적 상호작용과 집단지능

Hardware-Algorithm Co-Design for Emergent Behaviors in Modular Robotic Collectives

BARI Collaboration Meeting

BEI Robotics Lab.

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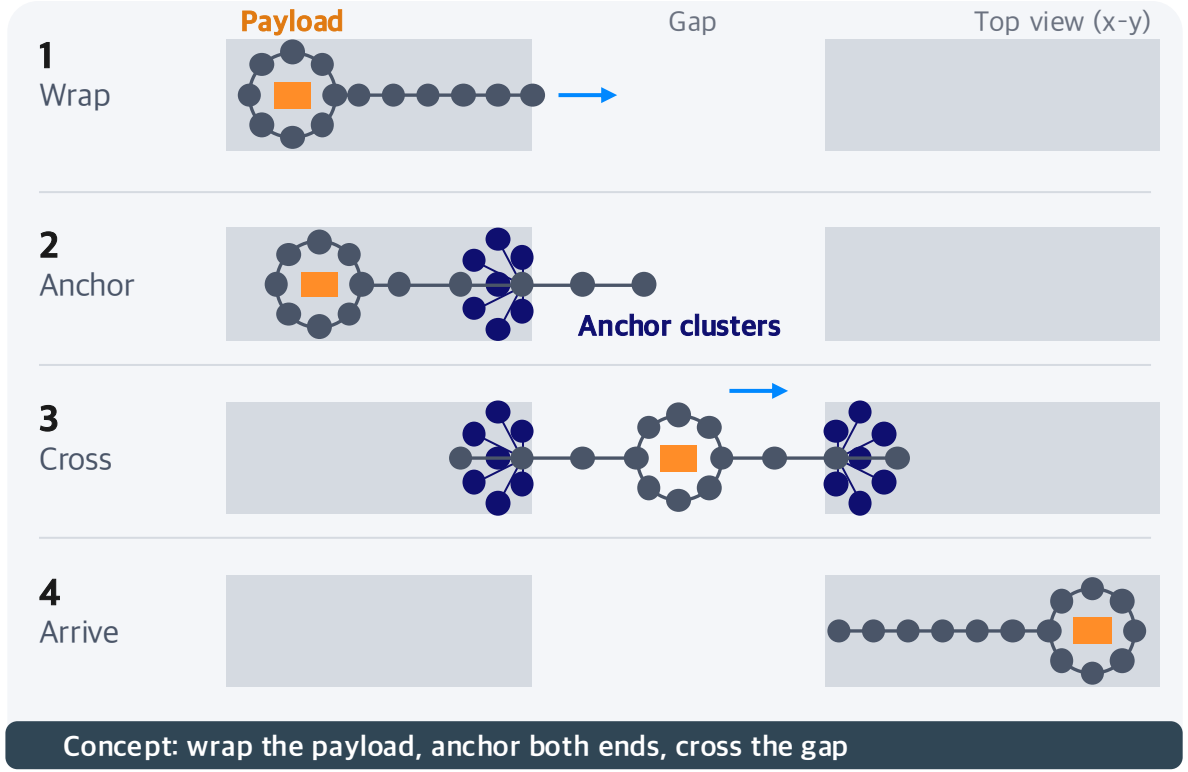
2026.10.19 Meeting Agenda

01	**Demo Scenario**	첫 target behavior / demo scenario
02	**Initial Platform**	운동 차원, 이동 방식, 사용하는 센서 등
03	**Interlocking**	Interlocking Mechanism Concept Proposal
04	**Next Deliverable**	연구실 별 역할, 다음 deliverable

Demo Scenario 결정 → 첫 번째 Prototype 제시 → Interlocking Mechanism 논의 → 공동 플랫폼 합의

Adaptive bridge formation: a living structural system

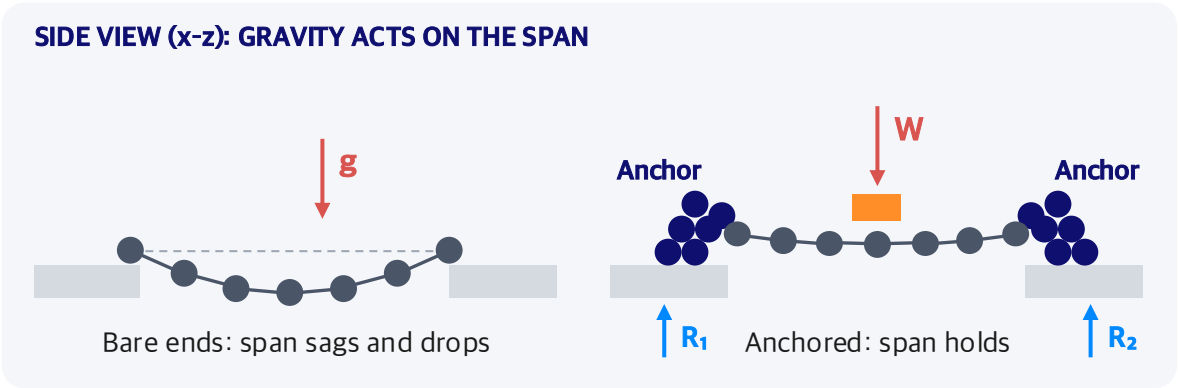
NATURE-INSPIRED REFERENCE



Concept: wrap the payload, anchor both ends, cross the gap

Army ant living bridges

- Ants adjust bridge length, width, and position as traffic and terrain geometry change.
- The structure balances path efficiency against the cost of allocating workers to the bridge.



Mechanical analysis targets

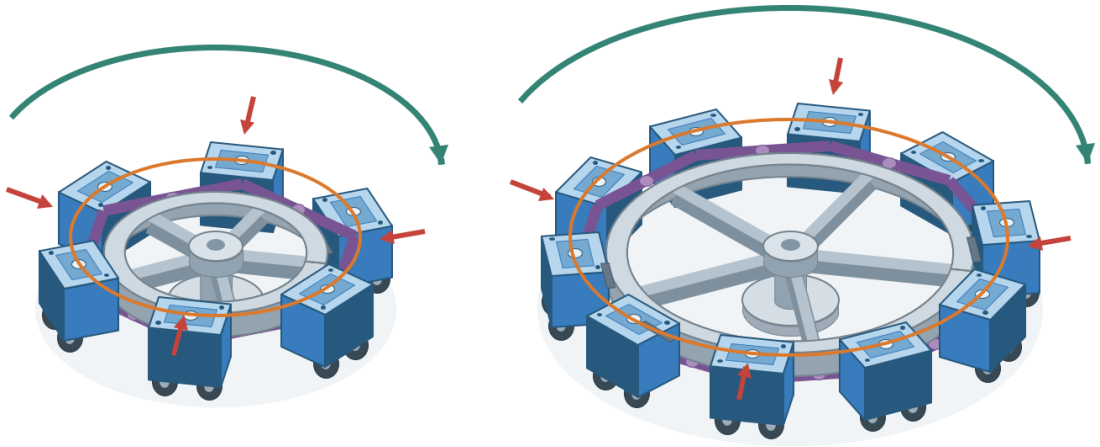
- Identify peak link force λ_{max}, global deflection δ_{max}, effective stiffness k_{eff}, and the onset of slip or contact loss under moving payloads.

Biological basis: Reid et al., "Army ants dynamically adjust living bridges in response to a cost–benefit trade-off," PNAS, 2015. DOI: 10.1073/pnas.1512241112

Valve Operation

Planar motion, but a three-dimensional force problem

NATURE-INSPIRED ANALOGY



Concept render: anchored modules rotating a floor-mounted valve

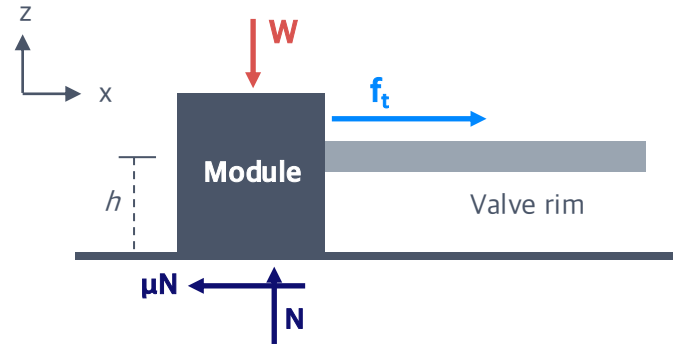
Motion is 2D

The rim turns in the floor plane; modules move only in x-y

Forces are 3D

Push height, floor grip and weight also act along z

Side view of one module



Pushing at height *h* can **tip** the module

Floor grip μN limits the push before it **slips**

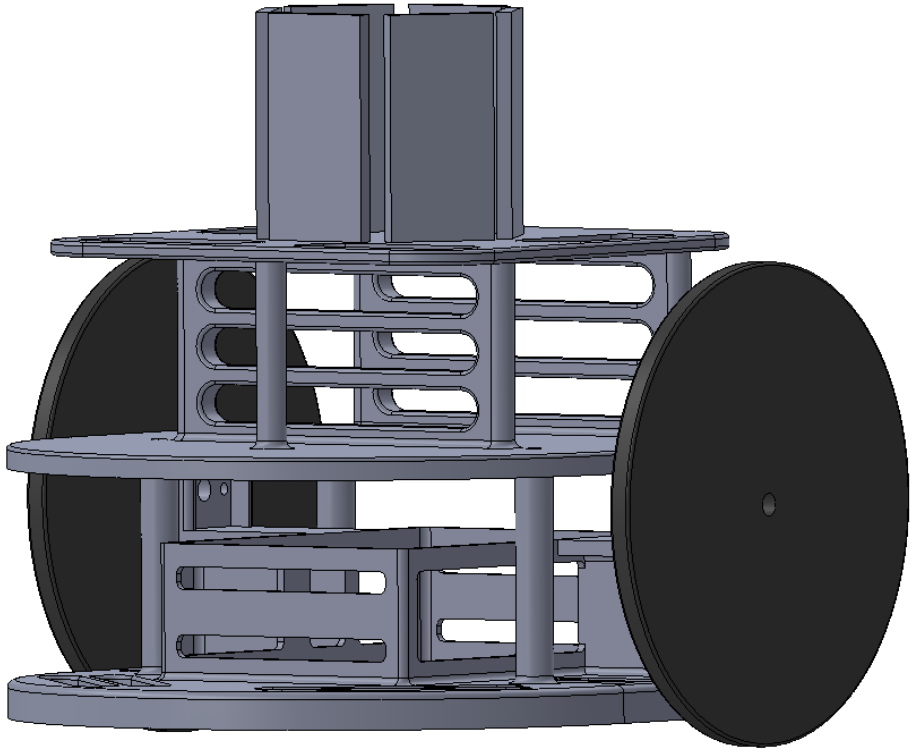
Still to analyze

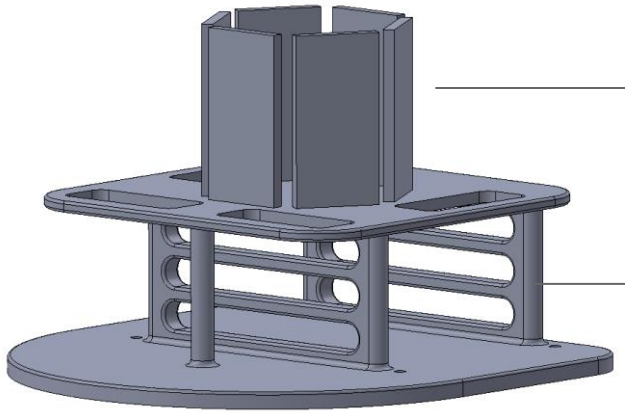
- How much push each module can give before it slips or tips
- How load spreads along the chain as the valve breaks free

Biological analogy: Wilson et al., "Multi-robot replication of ant collective towing behaviours," Royal Society Open Science, 2018. DOI: 10.1098/rsos.180409

Initial Platform Design

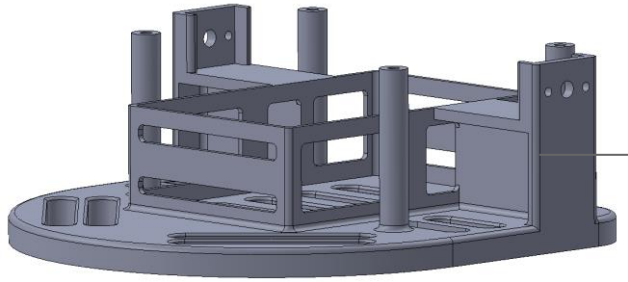
- **Size:** 90 × 103 × 93 mm
- **Purpose:** A first testbed to verify whether IR sensors alone can run the proposed algorithm reliably





3rd floor
IR sensor array
(IR – VL53L7CX)

2nd floor
Electronics mount



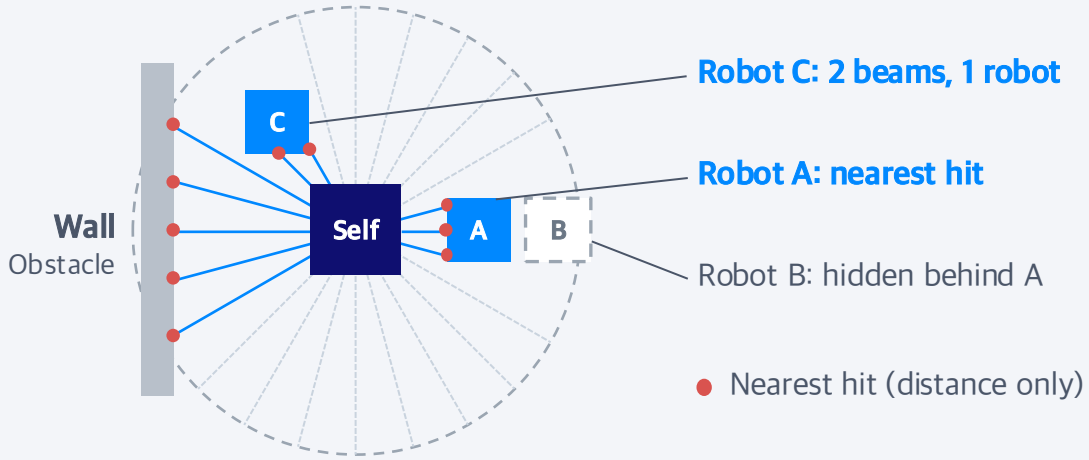
1st floor
Battery (mass balance), motors

Questions

QUESTION 1

With only the nearest distance per direction, how do we count robots and tell them from obstacles?

TOP VIEW: 360° IR RING, EACH BEAM RETURNS ITS NEAREST HIT



Counting robots

- One robot can span several beams
- A robot behind another is never measured

Robot or obstacle?

- A wall and a robot return the same kind of distance
- Nothing in the reading says what was hit

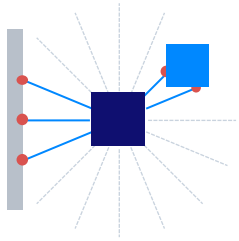
- The IR ring covers 360°, but each sensor reports only its nearest hit
- How do we turn these distances into an accurate neighbor count?
- How do we tell a robot from a wall or obstacle at the same distance?

Questions

QUESTION 2

Is the 360° IR ring enough, or should each module add a robot signal or a 2D LiDAR?

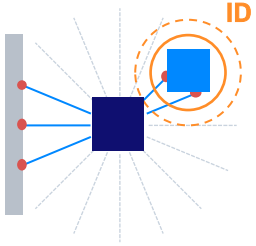
SENSING OPTIONS PER MODULE (TOP VIEW)



360° IR ring

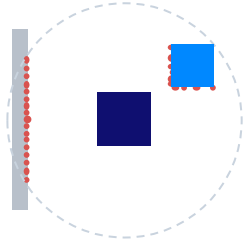
Nearest distance per direction only

Current



IR ring + robot signal

Neighbors broadcast an ID; walls stay silent



2D LiDAR

Outlines each object, but adds cost, size and power

QUESTION 3

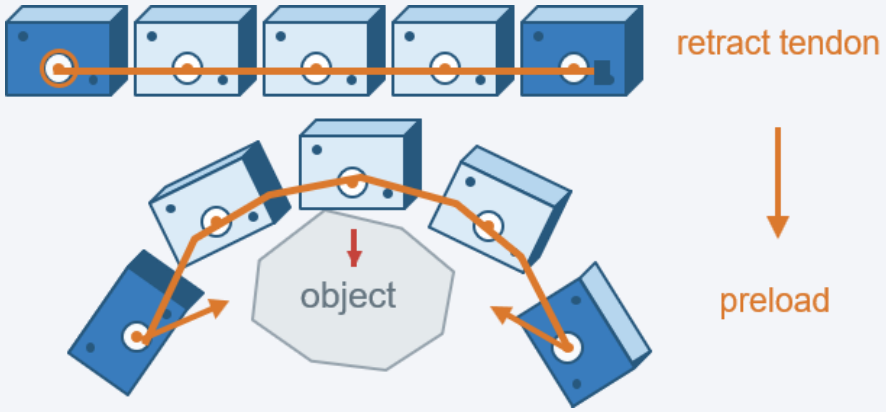
How much of the low-level control code should we provide?

- **Option A, driving only:** we provide motor control and odometry; your team builds sensing and the algorithm
- **Option B, driving + sensor readout:** we also provide IR distances from the 360° ring, ready to use
- **Open point:** who builds the neighbor count and density estimate on top of the raw IR readings?

Modular morphologies with shared-cable actuation

One cable locks the chain; module shape sets how many stable morphologies are possible

Interlocking by a shared cable



- **Builds on our cable robots:** one cable runs through all modules and locks the chain when retracted
- **Preload holds the shape** around the object, with no actuator per joint

Next: module shape sets the stable states

Cross-section, schematic



Cube	Hexagonal prism	Octagonal prism
3 joint states	5 joint states	7 joint states

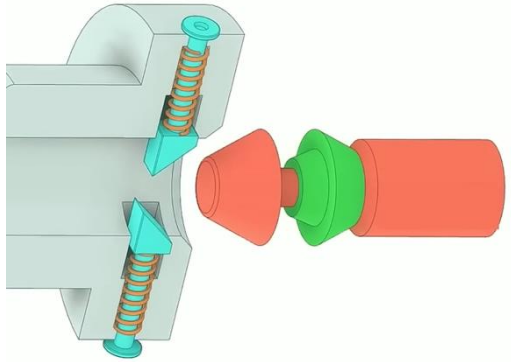
Blue: faces the next module can lock onto

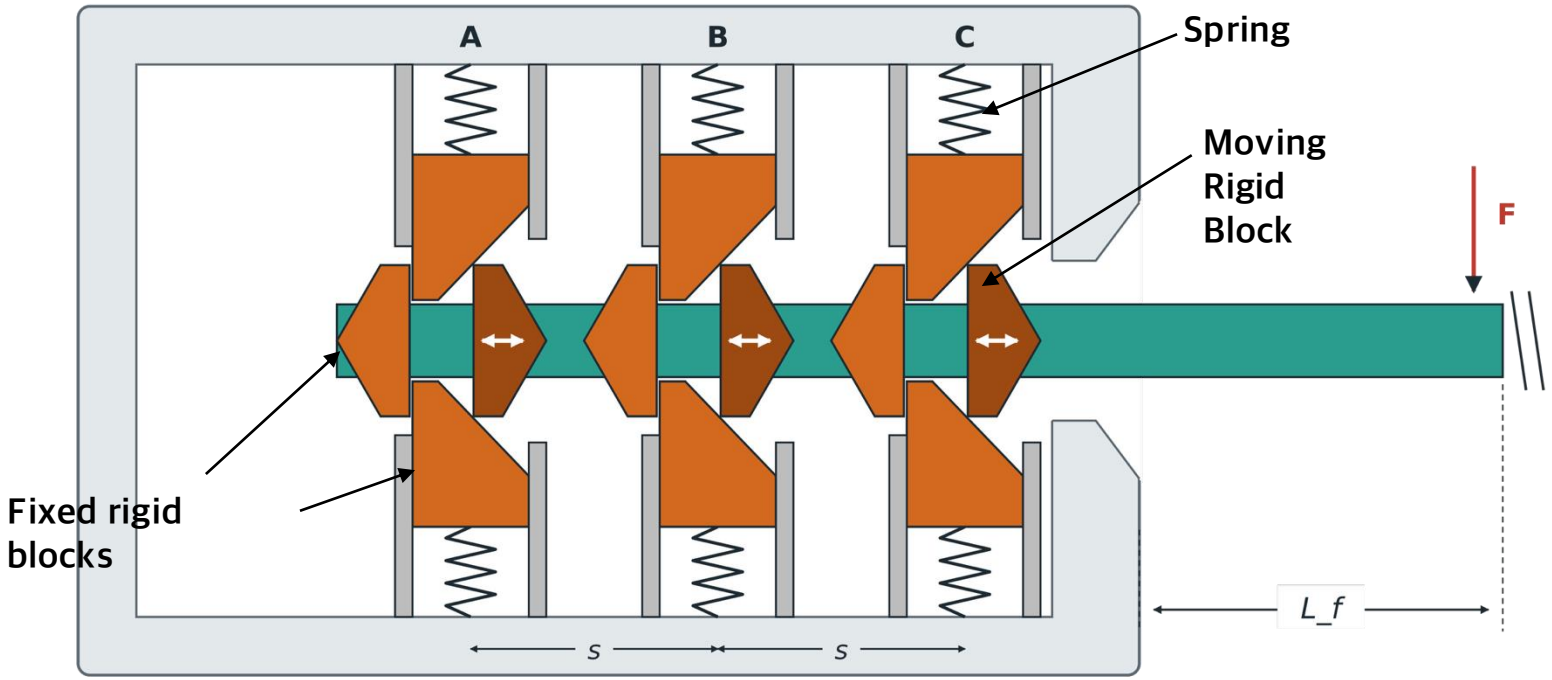
- **Hypothesis:** more lockable faces, more stable morphologies
- **Trade-off:** smaller faces, weaker lock
- **Next:** compare polyhedra to maximize stable states while keeping the lock strong

Concept Proposal II

Spring-Loaded Clamp Interlock

Passive Lock - Passive Release





Passive Lock - Passive Release

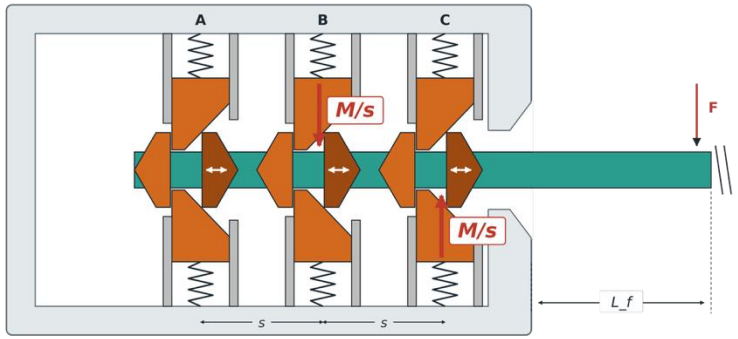
1. **Selectable docking depth**
   IR sensors measure the gap, so each module can choose how deep to dock

2. **Discrete stiffness control**
   Soft beam length sets how many clamps engage (A-C), so locking bending stiffness changes in steps

Spring-Loaded Clamp Interlock

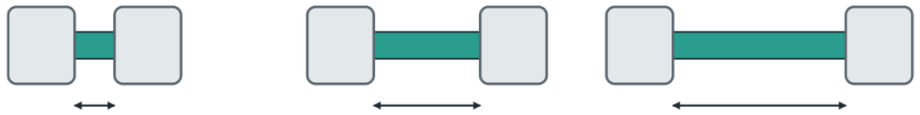
Main Design Parameters

① Clamp spacing s & clamp count



- **Clamp spacing s and clamp count set the coupling force, which determines bending stiffness.**

② Beam length vs. engaged clamps (3 clamps)

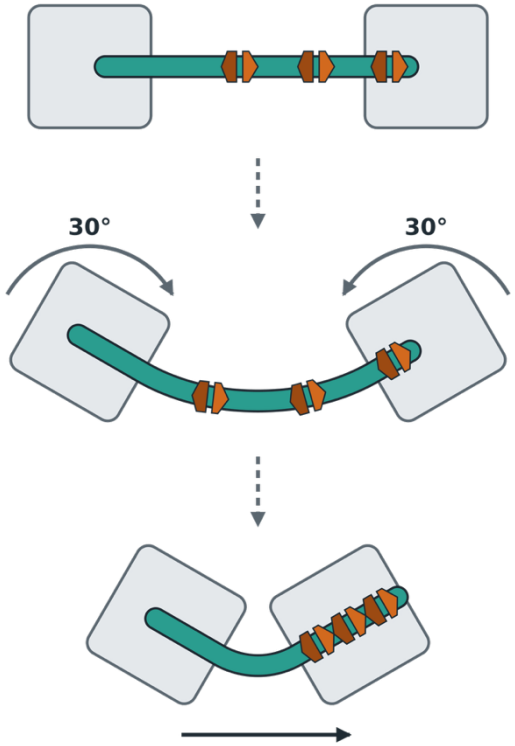


Short (A·B·C)
High stiffness

Medium (B·C)
Medium stiffness

Long (C)
Low stiffness

③ Engage → rotate → lock sequence



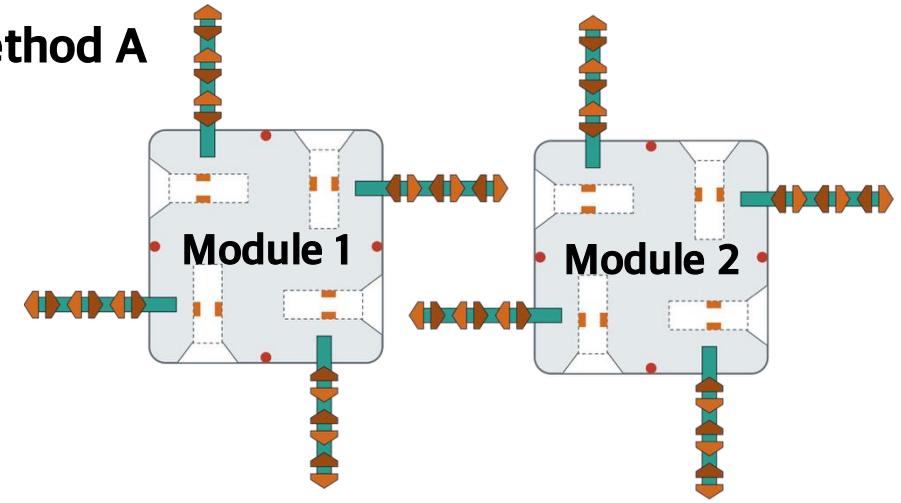
1. Passive soft interlocking

2. Easy rotation by low stiffness

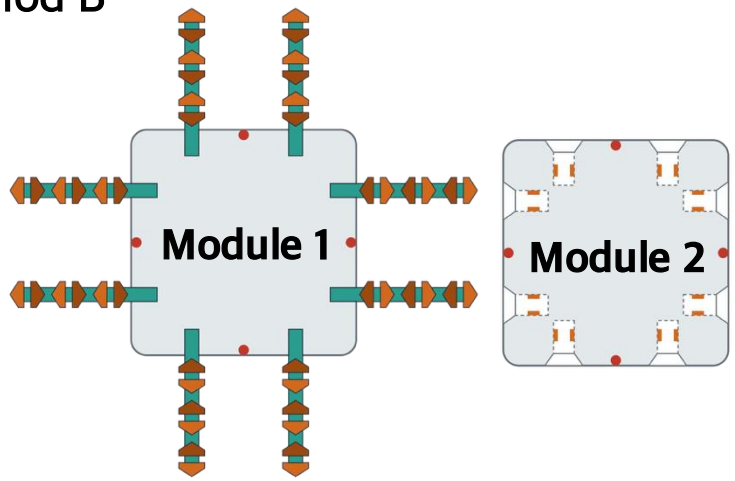
3. Passive hard interlocking for shape stability

Two Module Layouts for Beam-Receptacle Interlocking

- **Method A**



- Method B



Soft beam
TPU rod with 3 collar sets, protruding outward

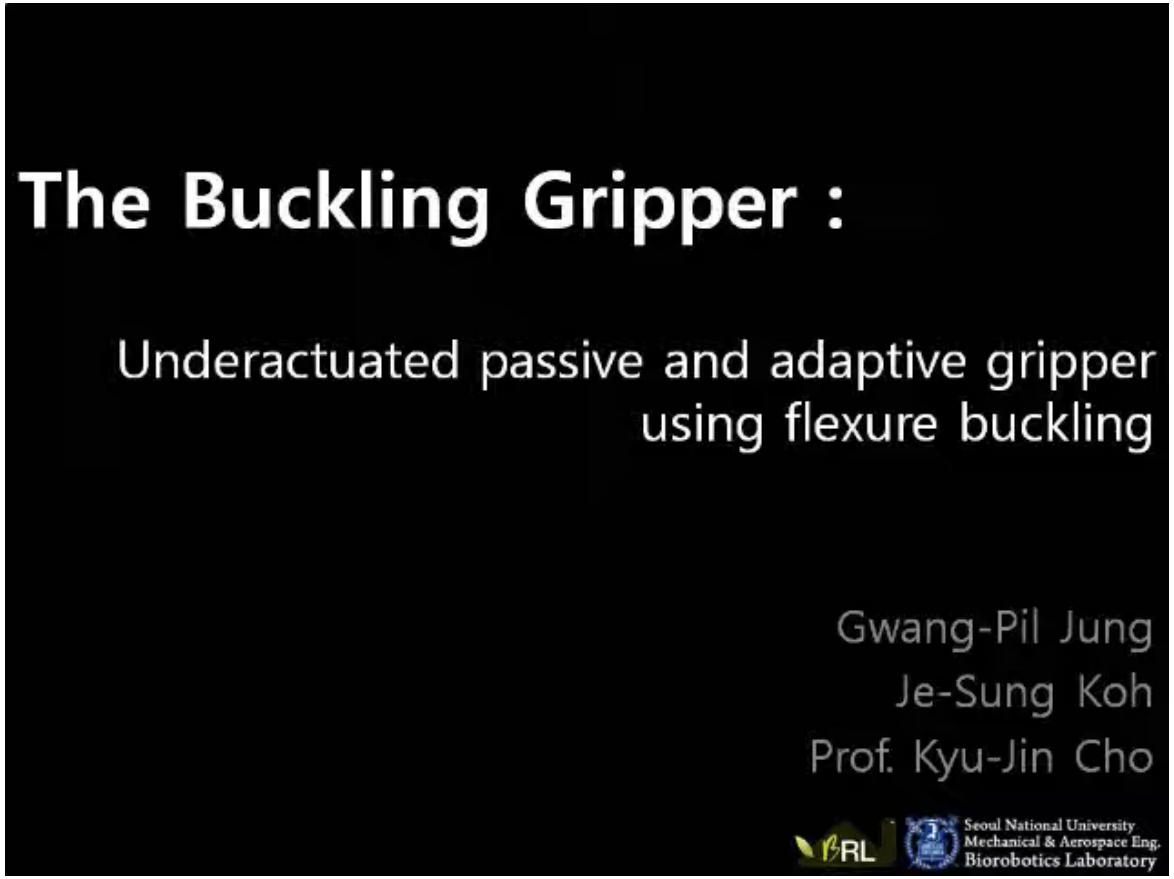
Receptacle
Funnel entry with internal wedge latch

IR sensor
Face center; measures distance to the opposing face

- The best layout depends on the polyhedron shape
- Soft beam placement on each module also matters
  - Choose the placement that keeps coupling control simple

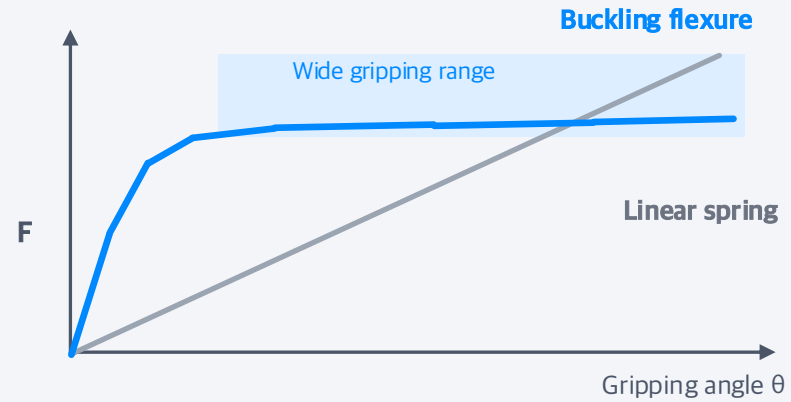
Flexural Buckling Gripper

Previous Work: Jung et al., IEEE Transactions on Robotics (2013)



In the video: each flexure buckles differently, so every spine grips the surface

Why buckling: near-constant force



Schematic, after Fig. 5 of the paper

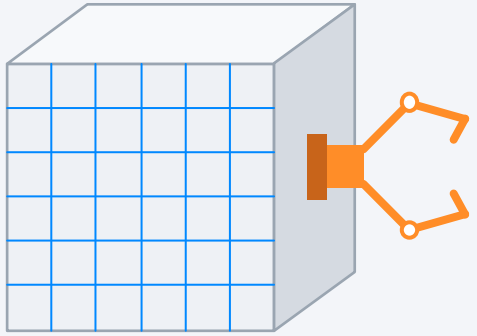
- After buckling, force stays **nearly flat**
- Even load sharing, **no differential** needed

Source: G.-P. Jung, J.-S. Koh, K.-J. Cho, "Underactuated Adaptive Gripper Using Flexural Buckling," IEEE Trans. Robotics, 29(6), 2013. DOI: 10.1109/TRO.2013.2273842

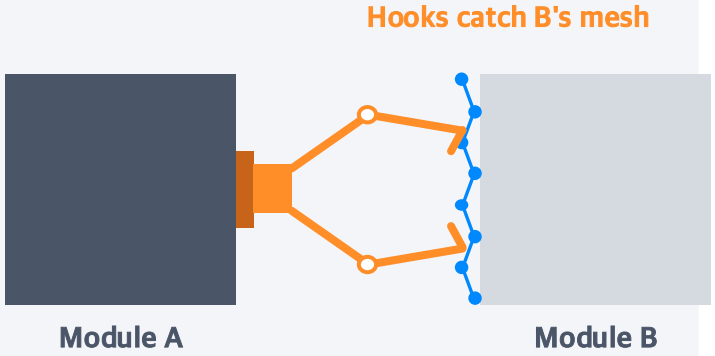
Interlocking Mechanism Using Buckling Gripper

One buckling gripper per module: it locks onto a neighbor's mesh side and grasps objects

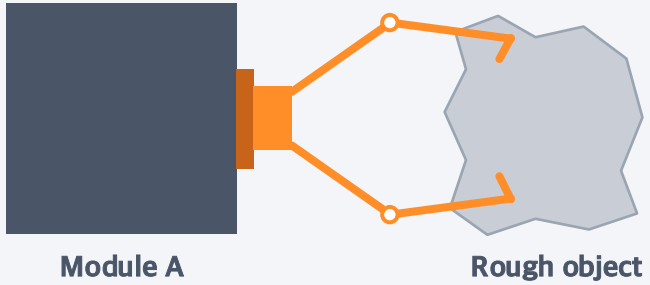
① Module design



② Lock to a neighbor



③ Grasp an object



Identical modules

Every module has one gripper and mesh on its other sides, so any module can lock onto any other

Module-to-module lock

Buckling spines adapt to the mesh, so faces need no precise alignment

Object grasp

The same gripper holds rough, irregular objects, as in Jung et al.

Gripper after Jung et al., "Underactuated Adaptive Gripper Using Flexural Buckling," IEEE T-RO, 2013. Figures are schematic.

APPENDIX F

BRL’s Previous Research on Multi-Robot Collaboration

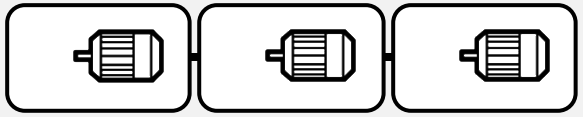
Force-Synergistic Modular Robot through Parallel Cable Actuation

Motivation: 결합을 통한 이득을 형상에서 힘으로 확장

Motivation

Kinematics

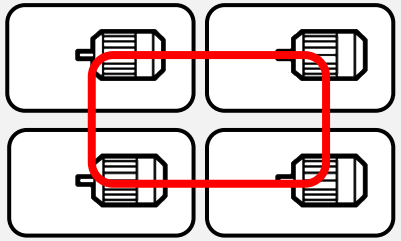
Multiple small robots - weak motors



Increase scale by connecting multiple robots

One large robot
- Powerful motor

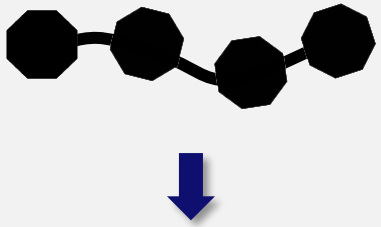




Dynamics

How to increase total force with multiple weak motors?

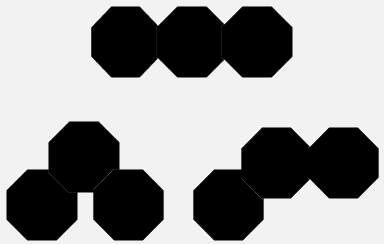
Idea

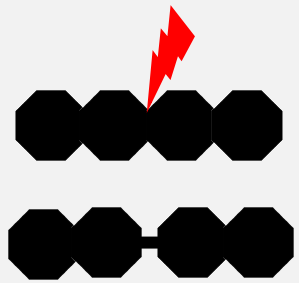


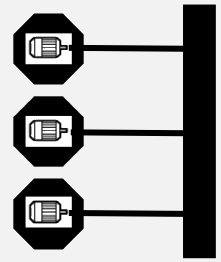
Various morphology

Robust connecting

Synergy of actuation







Cable location & Structural Shape

Enhanced chain integrity with a continuous cable

Use multiple cable (Tension multiplication)

Force-Synergistic Modular Robot through Parallel Cable Actuation

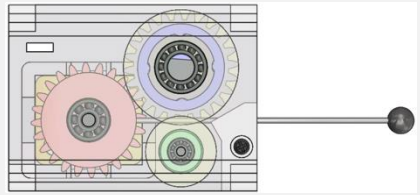
Design: 여러 모듈을 관통·연결하는 Cable Sending & Locking Mechanism

Connection

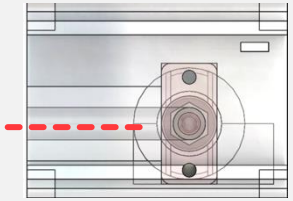
Cable sending module
(Slack enabling actuator)

Wire guide module

Locking module









Upper Bending



Straight



Lower Bending

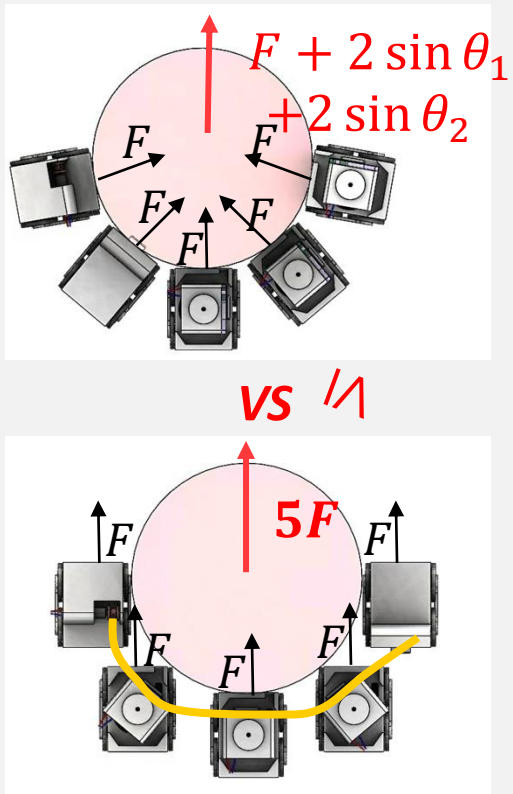


다음 연구로의 전환: Compliant Cable의 정밀 정렬의 어려움. Connection Failure 多

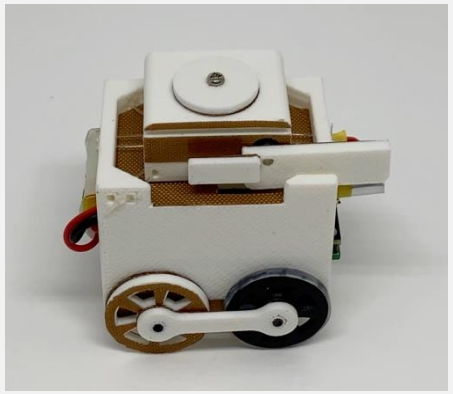
A passively Guided Cable-Connected Robot Team

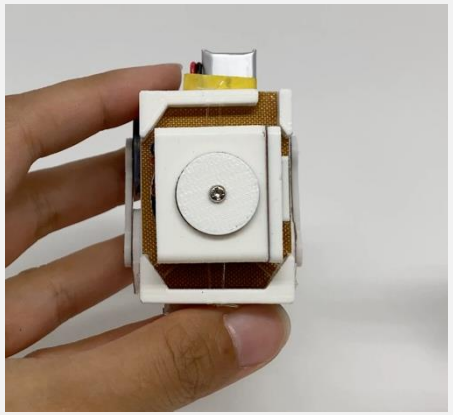
Motivation: Cable로 기연결된 로봇 + Cable compliance로 물체 형상에 수동 적응하고 추진력 손실을 줄임

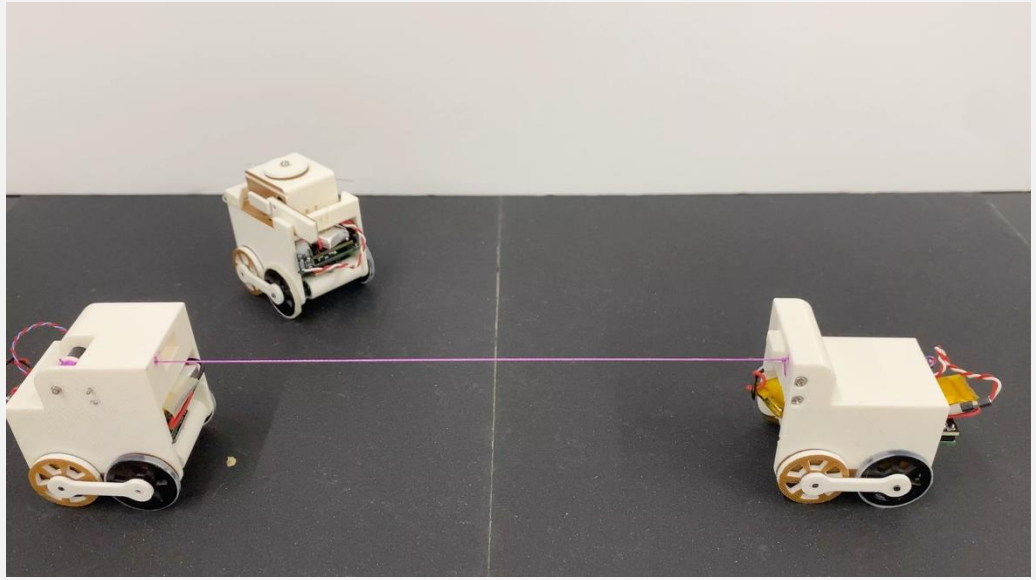
Motivation



Connection / Passively Rotated Guide





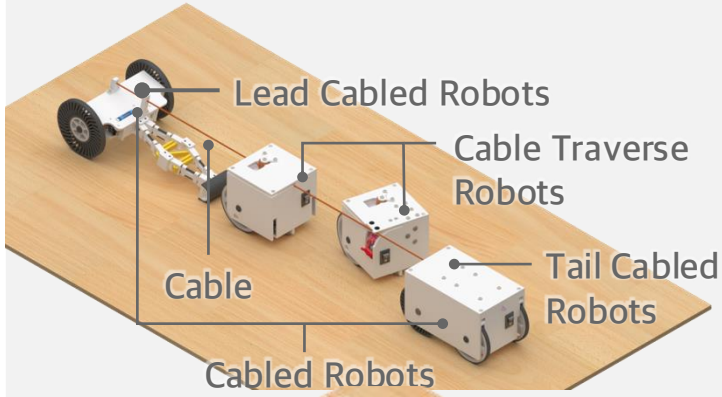


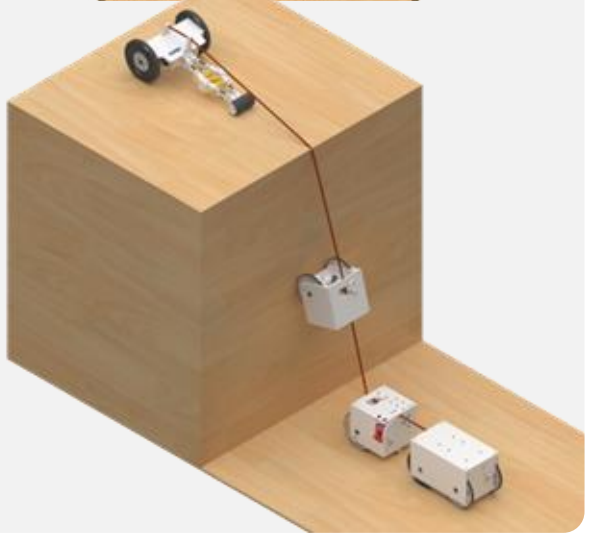
다음 연구로의 전환: Planar 움직임 한계 ⇒ 3차원으로의 확장 필요

Cable-Mediated Physical Collaboration of Heterogeneous Swarm

Design: 경로 형성 로봇과 케이블 추종 로봇의 역할 분리 • 구조적/동역학적 결합 상태를 선택적으로 전환

System Architecture

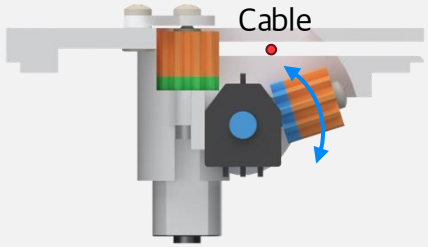


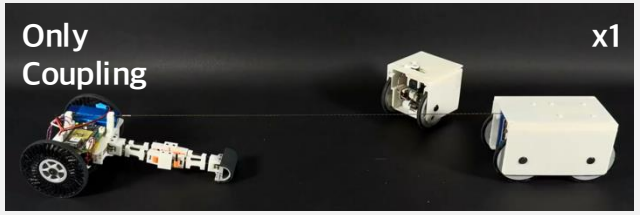


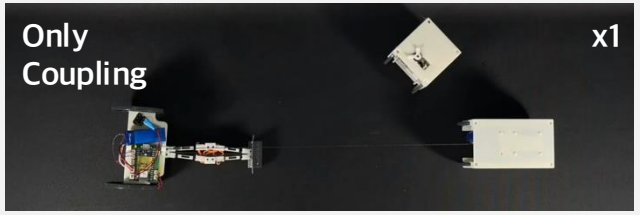
Connection Mechanism

Coupling

Coupling State Control & In-Cable Traversal

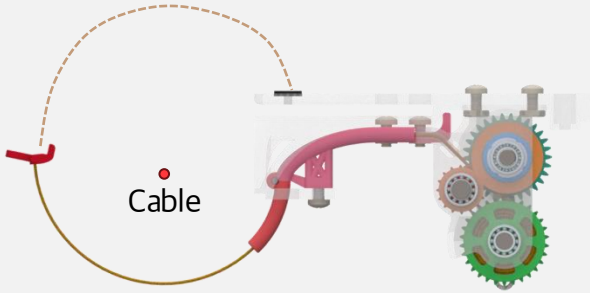


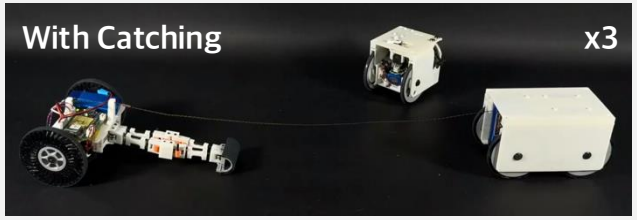


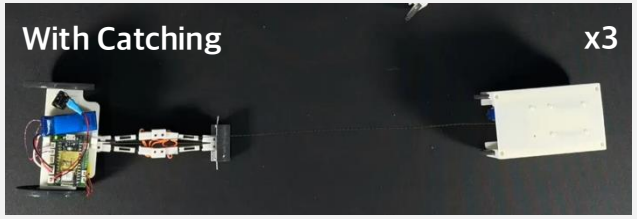


Catching

Wide-range alignment range







Cable-Mediated Physical Collaboration of Heterogeneous Swarm

Result: 케이블 추종과 coupling-state control을 통한 다양한 장애물 극복

Obstacle Overcoming by Cable Traversal







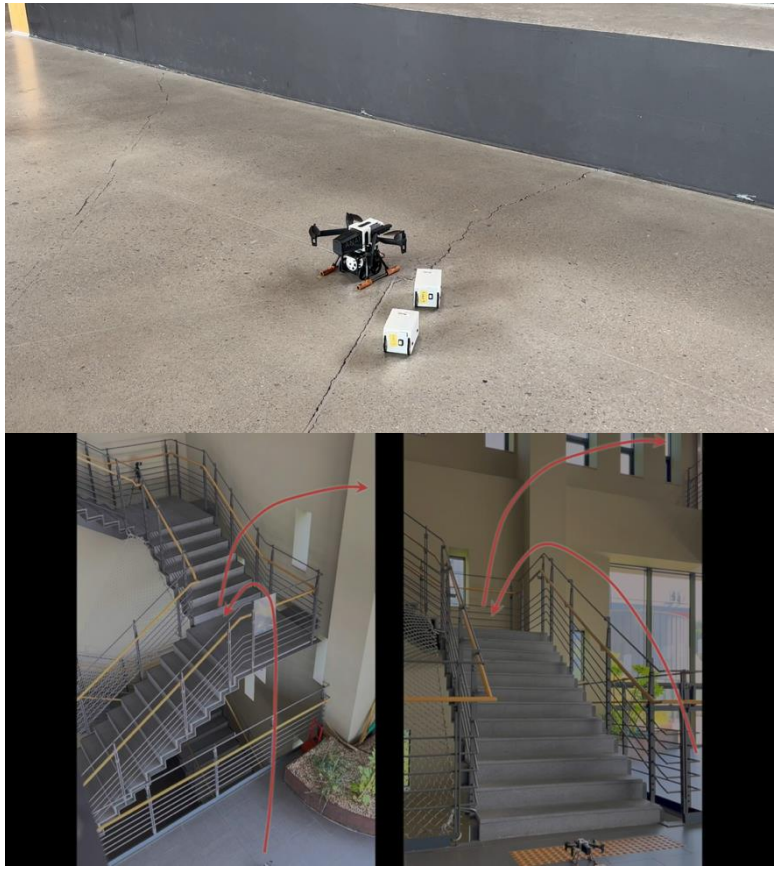
Inertial Decoupling







Demonstration



다음 연구로의 확장: Closed-loop 결합과 local interaction을 통한 adaptive & collective functionality